01 · LESSON

# Probability and statistics

*9A Genetics and evolution · Lesson 3 of 7*

Date _____ Class _____

**Today I can 今天我能**

- ☐ I can explain what probability is and write a probability as a percentage, a fraction and a decimal.
- ☐ I can calculate an experimental probability from results and compare it with a theoretical probability.

**What happens today**

1. Do Now: chance of rain
2. Learn: What statistics is for · Probability · and more
3. Practical: Investigation: two dice
4. Challenge, then the exit ticket

Student Book pages 10 to 11 · Workbook pages 7

**What I learned today (one or two sentences) 今天我学到了：**



02 · KEY VOCABULARY

# Key Vocabulary

| Term | 中文 | Definition |
|---|---|---|
| **statistics** | 统计 | collecting, analysing and presenting data |
| **sample** | 样本 | a small part of a group that is measured to estimate the whole group |
| **mean** | 平均数 | the sum of the values divided by the number of values |
| **median** | 中位数 | the middle value when the values are in order |
| **mode** | 众数 | the most common value in a set |
| **probability** | 概率 | the chance of something happening |
| **percentage** | 百分比 | a number out of 100, shown with the % sign |
| **fraction** | 分数 | a part of a whole, written as one number over another, e.g. 1/4 |
| **decimal** | 小数 | a number written with a decimal point, e.g. 0.25 |
| **experimental probability** | 实验概率 | a probability worked out from the results of an experiment |
| **theoretical probability** | 理论概率 | a probability worked out by reasoning about all the possible outcomes |

**Command words 指令词:** Calculate 计算 · Complete 完成（填写） · Draw 画出 · Explain 解释 · Fill in 填写 · Give 给出 · Match 连线（配对） · Write 写出



03 · QUICK REVIEW

# Quick Review

*chance of rain*

A weather app says there is a 76% chance of rain between 11:00 and 12:00 tomorrow.

**1** What do you think 76% means here?

**2** Write 76% as a fraction and as a decimal.

**3** Is rain certain? How do you know?



04 · CORE

# Core 1 of 3

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**1** Match each scientific term to its definition. Draw a line.

| | |
|---|---|
| mean | the most common value in a set |
| median | the chance of something happening |
| mode | the sum of the values divided by the number of values |
| probability | the middle value in a set of values in order |

**2** Complete the probability scale. Fill in the missing values.

| Percentage | Fraction | Decimal | Word |
|---|---|---|---|
| 0% | 0 | 0 | impossible |
| 25% | 1/4 | | |
| 50% | | 0.5 | even chance |
| 75% | | | likely |
| 100% | 1 | | certain |

# Core 2 of 3

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**3** Two dice investigation. Throw two dice, add the numbers, and tally the score. Do 50 throws.

**a)** Complete the table. Experimental probability = frequency divided by 50.

| Score | Tally | Frequency | Fraction (out of 50) | Decimal | Percentage |
|---|---|---|---|---|---|
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | | |
| 9 | | | | | |
| 10 | | | | | |
| 11 | | | | | |
| 12 | | | | | |





# Core 3 of 3

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**3** *continued*

**b)** Which score was the mode (the most common) in your results?





# Challenge 1 of 2

*When Core is finished. Full sentences. 完成基础题后做；请写完整句子。*

**C1** Achondroplasia is an inherited condition. If neither parent has it, the theoretical probability of a child having it is 0.003%. If one parent has it, the probability is 50%. If both parents have it, the probability is 75%.

**a)** Write the probability for a child with one affected parent as a decimal and as a fraction.

**b)** A couple (one parent affected) has two children without the condition. Explain whether their third child is more likely to have it.





# Challenge 2 of 2

*When Core is finished. Full sentences. 完成基础题后做；请写完整句子。*

**C1** *continued*

**c)** Explain why these are theoretical probabilities and not experimental probabilities.

**C2** Calculate the theoretical probability of throwing a number less than three with one dice. Give your answer as a fraction, a decimal and a percentage. Then explain why an experimental probability from 100 throws is more reliable than one from 10 throws.

# Exit Ticket

*Hand this card in at the end of the lesson.*

Name _____

**1** What is probability?

**2** Write 5% as a decimal.

**3** A bag has 20 red, 10 green, 5 blue and 15 yellow beads. Give the probability of taking a blue bead as a fraction and as a percentage.

**Today I can:** Goal 1 ☐ Goal 2 ☐